The Goat Herd

A Preprint

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Abstract

A one-line human brief — reservoir computing, with a goat *herd* as the metaphor — became a fragmentation study: cut the house 1024-unit sparse tanh reservoir's mixing matrix into H independent same-total-width blocks, per-block spectral radius matched to the intact pond (ρ=0.95), readout always seeing the full 1024-wide concatenation. We registered a fragmentation *tax*; the pre-named alternative fired instead and, in its final ridge-first wording, hit two consecutive registered passes — ten seeds across four registered passes are on record. Two findings. (1) **Fragmentation is a free lunch, ridge-first**: the 64-block herd never costs ridge validation bpc — strictly negative in all 10 seeds — and wins outright on the trained logistic instrument in 8 of 10, the two misses being dead ties (+0.0004, +0.0006), not reversals. The house champion at ρ=0.95 sits measurably past its own bpc optimum; block-diagonalizing the mixing matrix cools the effective operating point for free. (2) **Memory survives**: the herd's decode depth never falls more than 0.20 characters short of the intact pond's, 10/10 seeds, despite each block seeing only 16 of the 1024 neurons. A third, stronger-sounding claim — that a weakly coupled ("flocked") herd beats the intact pond outright — sits at 9 wins in 10 seeds and is deliberately *not* promoted: it missed its registered all-seeds bar, and unpromoted claims are reported here as clearly-labeled context only.

Keywords reservoir computing · echo state networks · block-diagonal recurrence · modularity · spectral radius · pre-registration

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§1 The question

The lab's standard machine is one large well-mixed pond: a sparse random recurrent matrix, spectral radius ρ=0.95, every unit potentially entangled with every other within a few steps [1]. The herd brief asks the opposite: what if the same 1024 neurons were instead 64 *separate* ponds of 16 neurons each, grazing the same text side by side, never mixing? At matched total readout width, is fragmentation a tax (less mixing, less computation), a free change of coordinates, or — the outcome no registered prediction favored — an improvement?

§2 The machine

The house sparse tanh reservoir (N=1024, fanin 10, house input map and bias, leak 1.0), with its mixing matrix W made block-diagonal: H equal contiguous blocks of n = 1024/H units, each block's own spectral radius set to 0.95 — the intact pond's value — so per-block gain is matched, not diluted. The readout always sees the full 1024-wide concatenated state, eliminating the feature-width confound the lab measured separately. Three instruments carry the paper: ridge validation bpc (optimizer-free), the trained logistic readout's validation bpc, and U3 decode depth — how many characters of recent history a ridge probe can reconstruct — all linear-readout instrumentation in the reservoir-computing tradition [3]. Standard text8 splits [4], 2M train characters.

The first pass (seed 0) swept H ∈ {1, 4, 16, 64}, plus two variants: *flocked* herds, where every goat also receives its input map applied to the herd's mean state — a weak mean-field "bellwether" channel, the coupling scaled to spectral radius 0.2 and the whole matrix rescaled so total gain matches the independent twin (topology changes, gain does not) — and a *temperament-diverse* herd at H=16 (per-goat ρ log-spaced 0.80 → 1.128, geometric mean 0.95). A gate ran first: the H=1 cell through the copied pipeline reproduced the archived champion pond of the addressed-pond study to Δ = 0.0000 on all three instruments, so every herd cell is directly comparable to the lab's archive.

§3 What we registered in advance

Seed 0 registered five ordinals with named alternatives. The registered tax prediction missed onto its pre-named alternative, and the registered depth band missed from above — in the profitable direction, both times — and those two misses are the paper:

- **R1** (fragmentation tax): bpc strictly increases along H = 1 → 4 → 16 → 64. Named alternative **A1** (cooling win): fragmentation *decreases* bpc toward the pond's ρ=0.8 ladder optimum. (A1 fired.)
- **R2** (flocking rescue): the bellwether channel helps at H=16 and H=64. Named alternatives: A2, a readout-suffices null (|Δ| ≤ 0.005 everywhere); A3, groupthink (coupling hurts). (A2 fired at seed 0 — and was itself refuted by the reruns; see §4.)
- **R3** (diversity no-win): the temperament portfolio does not beat the uniform herd. (Hit; the portfolio cost +0.0070.)
- **R4** (depth falls): decode depth strictly decreases with H. Named alternative **A5** (redundant-encoding rescue): depth(H=64) lands within 1.0 character of intact. (The ordinal hit — and the magnitude belonged to A5: total loss 0.20 characters, against a registered band of [4, 8] characters of *remaining* depth that the herd beat from above.)
- **R5** (hot goats decode deeper): the diverse herd out-decodes the uniform herd at H=16. (Hit, thinly: +0.07 characters.)

Three registered replication passes followed — seeds 1–3, 4–6, and 7–9 — each written before running. The middle pass bundled its claims conjunctively and paid for it: its flock sub-claim missed while the composite, the cooling win, and the depth rescue each hit clean, and the registered rule blocked *all* promotion that pass. The lab journaled the bundling as the registration's mistake and re-registered the surviving claims *unbundled* for seeds 7–9, each with its own promotion rule: T1, the flocked-herd composite (all 3 seeds beyond −0.005); T2, the cooling win in ridge-first wording (ridge win in all 3 seeds AND logistic beyond −0.005 in ≥2 of 3); T3, the depth rescue (within 1.0 character in all 3 seeds).

§4 What happened

Seed 0's ladder set the shape: logistic val bpc 2.6727 → 2.6742 → 2.6627 → 2.6586 along H = 1/4/16/64 (the H=4 step inside the ±0.005 tie band; H=16 and H=64 clear wins), with the optimizer-free ridge instrument agreeing in direction and size (3.1472 → 3.1303/3.1321 at H=16/64). The registered tax band ([+0.01, +0.10] bits) was missed with the opposite sign: the registered tax is a measured profit. Ten seeds later, the pattern of Table 1 is the finding.

Table 1. Intact pond (H=1) vs the 64-block independent herd, all ten seeds. Δ = herd − intact (negative favors the herd); depth in characters. Runs: ledger #12 (seed 0), #13 (1–3), #14 (4–6), #15 (7–9). Zero AUTO-VOIDs in 48/48 cells across the family.

seed	Δ ridge val	Δ logistic val	depth H1	depth H64	depth loss
0	−0.0151	−0.0141	10.11	9.91	0.20
1	−0.0140	−0.0115	10.10	9.94	0.16
2	−0.0209	−0.0124	10.12	10.03	0.09
3	−0.0059	+0.0004	10.07	9.91	0.16
4	−0.0181	−0.0055	10.07	9.95	0.12
5	−0.0080	−0.0084	10.06	10.06	0.00
6	−0.0223	−0.0197	10.11	9.94	0.17
7	−0.0140	−0.0084	10.14	10.03	0.11
8	−0.0023	+0.0006	10.05	10.04	0.01
9	−0.0217	−0.0180	10.10	10.01	0.09

The ridge column is negative ten times out of ten. The logistic column is beyond the −0.005 bar eight times out of ten, and the two misses (seeds 3 and 8) are dead ties a fraction of a thousandth of a bit wide — the herd never loses, it occasionally only draws. The depth column's worst case is 0.20 characters, on blocks that each see 16 neurons where the intact pond's probe sees 1024. T2 and T3 each hit their unbundled registered bars on seeds 7–9 and promoted on their own merits (T2's second ridge-first pass; T3's third pass).

What did not survive, reported with equal weight. The flocked-herd composite — H=64 with the bellwether channel beating the *intact* pond — hit its bar on seeds 4–6 (−0.0132 / −0.0166 / −0.0152) but missed on seeds 7–9: −0.0095 / −0.0024 / −0.0277, where seed 8's −0.0024 sits inside the tie band. All-time the composite is 9/10, the sole miss a tie, and it is *not in the lab's findings book* — the registered rule needed all three, and a 9/10 regularity that failed its stated bar stays unpromoted. Likewise the flock sub-benefit (flocked vs its own independent twin): favored in 11 of 14 comparisons across seeds 0–6, mean ≈ −0.003 bits, sign-unstable — retired to measured-context wording, deliberately not re-registered. And the seed-0 registered claims R3/R5 (diversity costs bpc +0.0070; hot goats decode +0.07 deeper) remain single-pass results, not findings.

§5 What it means

The champion is running slightly hot. The intact pond at ρ=0.95 sits measurably past its own bpc optimum (the lab's ladder places that optimum near ρ=0.8). Cutting W into 64 blocks — while explicitly holding per-block spectral radius at 0.95 — mixes less, and behaves as an effectively *cooler* operating point at matched width and matched per-block gain. The win is not an artifact of the trained instrument's optimizer: the optimizer-free ridge ruler shows it in all ten seeds, which is why the promoted wording is ridge-first. Decomposing a reservoir into decoupled sub-reservoirs has precedent as an architecture proposal [2]; what this study adds is the matched-gain, matched-width control and the pre-registered accounting that makes "free lunch" a measured statement rather than a design preference.

Redundant partial views collectively remember. Sixty-four 16-dimensional glimpses of the same character stream are almost as ridge-decodable, together, as one clean 1024-dimensional encoding — worst loss a fifth of a character across ten seeds. Depth, unlike bpc, was expected to pay heavily for fragmentation (the registered band put the herd's remaining depth at just [4, 8] characters); it barely paid at all.

And a discipline note the lab is keeping. The strongest-sounding herd sentence — "the flocked herd beat the big pond every time" — was true for eight consecutive seeds, then drew on the ninth; nine wins in ten is a humbler sentence than the seven-for-seven the claim stood at when its final registered pass was written. The program's rule against promoting on accumulated enthusiasm is exactly what kept it out of this paper's findings, and we record it here as context, clearly labeled, because that is what the section exists for.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. Y. Xue, L. Yang, S. Haykin, "Decoupled echo state networks with lateral inhibition," *Neural Networks* 20(3), 2007.
3. M. Lukoševičius, H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review* 3(3), 2009.
4. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **77cd2338** — `2026-08-11-caity-goat-iii`: the herd brief; 9 cells, seed 0 (run ledger #12). Registered before running; exact-reproduction gate on the archived champion passed at Δ = 0.0000.
- **8f5bdd07** — `2026-08-11-goat-herd-rerun1`: 15 cells, seeds 1–3 (run ledger #13). The replication split: cooling win 2/3 on logistic, 3/3 on ridge; the flock null refuted.
- **cc447b8d** — `2026-08-11-goat-herd-rerun2`: 15 cells, seeds 4–6 (run ledger #14). All but one registered claim hit; the conjunctive promotion rule blocked promotion, journaled as a registration-design error.

- **d3881e88** — `2026-08-11-goat-herd-rerun3`: 9 cells, seeds 7–9 (run ledger #15). Claims unbundled; T2 (cooling win, ridge-first) and T3 (depth rescue) promoted on their own rules; T1 (flocked composite) hit 2 of 3 seeds, short of its registered all-3 bar, and stays out.

All claims judged strictly against the registered wording, misses included above; 10 seeds across four registered passes; replicated before publication.
